

# PRINCE SAMUEL KYEREMANTENG

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## EDUCATION

### Lund University, Sweden

Exchange Program; Master's in Computer Science  
Erasmus+ Scholar; **About €6000 in grant.**

Lund, Sweden

**Jan – Jun 2026**

### University of Ghana, Ghana

MPhil Computer Science; **CGPA 3.75/4.0**

Accra, Ghana

**Oct 2024 – Sep 2026**

**Thesis:** Gap-Geometry-Driven Imputation of IoT Temperature Time Series: Deterministic Priors with Validation-Gated Neural Correction

### University of Ghana, Ghana

Bachelor's in Computer Science; **CGPA 3.45/4.0**

Accra, Ghana

**Oct 2019 – Oct 2023**

**Thesis/Project:** SmartFlow: IoT-Based Smart Irrigation and Environmental Monitoring System

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## SKILLS SUMMARY

- **Languages:** Python, SQL, JavaScript, Dart, HTML, CSS, Tailwind
- **Frameworks:** Django, FastAPI, Flutter, PyTorch, TensorFlow, Pandas, ScikitLearn, React
- **Tools:** Git, Postman, PostgreSQL, PgAdmin, LLM, Agentic AI, RAG, Docker, REST API
- **Platforms:** GitHub, Jupyter Notebook, VSCode, Codex, Linux Servers, Overleaf
- **Soft Skills:** Project Management, People Management, Excellent Communication, Versatile

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## RESEARCH INTERESTS

- AI-Enabled Software Engineering (AI4SE, SE4AI, Agentic AI, etc.)
- Software Security and Secure Software Development
- Empirical Software Engineering/Development, Testing, and Optimization
- AI and Machine Learning Systems Engineering (LLMs, Agent Orchestration), Autonomous Systems
- Reliable machine learning for intelligent sensing, uncertainty-aware inferencing, and edge AI

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## RESEARCH EXPERIENCE AND PUBLICATIONS

### Research Experience

- Graduate Teaching and Research Assistant (TA/RA), University of Ghana 2023 – Present
- MPhil Thesis Researcher, University of Ghana Oct 2024 – Sep 2026

### (1) Book Chapter: Kyeremanteng, P. S., et al (2025).

Federated Learning at the Edge: Enhancing Privacy and Efficiency in Distributed AI Systems

**Publisher:** [IntechOpen](#) || <https://www.intechopen.com/chapters/1230198> || **Co-author**

### (2) A Review Manuscript: Kyeremanteng, P.S., et al.

A Systematic Review of Large Language Models, 2017 to 2026

**Publisher:** Preprints.org || **DOI:** [10.20944/preprints202607.0296.v1](https://doi.org/10.20944/preprints202607.0296.v1) || **First Author**

**(3) Short Paper (Technical Report): Kyeremanteng, P.S., et al.**

ClarifyFlow: A Clarification-Aware Agentic Pipeline for Code Generation.

**LINK:** <https://github.com/DevPrinceK/clarifyflow>

**(4) A Review Manuscript: Kyeremanteng, P.S., et al.**

Computational Intelligence for Missing Data Imputation: Techniques, Challenges, Innovations, and Applications - A Systematic Review

**Publisher:** International Journal of Computational Intelligence Systems || **Under Review** || **First Author**

**(5) Article Manuscript: Kyeremanteng, P.S., et al.**

Gap-Aware Prior-Guided Imputation with Conservative Residual LSTM-VAE Refinement for IoT Temperature Sensor Streams

**Publisher:** International Journal of Computational Intelligence Systems || **Under Review** || **First Author**

**(6) Manuscript: Kyeremanteng, P. S., et al (2026).**

PoAD-Lite: A Lightweight, Activity-Aware Consensus Protocol for Scalable and Energy-Efficient IoT Blockchains

**Publisher:** IET Smart Cities || **DOI:** <http://dx.doi.org/10.1049/smc2.70034> || **Co-Author**

**(7) Manuscript: Kyeremanteng, P. S., et al.**

Secure Sharding in PoAD: Enhancing Scalability Without Security Trade-offs

**Publisher:** IET Blockchain || **Manuscript ID:** BLC-2025-08-0170 || **Under Review**

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## CONFERENCES AND PRESENTATIONS

**(1) Postgraduate Research Workshop, Computer Science Department – University of Ghana (2025)**

**Presentation:** Missing Data Imputation Techniques: Applications, Challenges, and Innovations

**Award:** 3rd Best Paper Award

**(2) Nyansapo AI Symposium on Sustainable Development**

Organized by the Ghana Atomic Energy Commission and Université Paris-Saclay (2023)

**Presentation:** AI-Based predictive system for electricity outage forecasting.

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## WORK EXPERIENCE

**Teaching and Research Assistant | [University of Ghana](#)**

**October 2023 – Present**

- Conducted tutorials on machine learning, heuristic search, and knowledge representation.
- Assisted in research in the labs of two professors during my time as a research assistant.
- Assisted in teaching Artificial Intelligence, database management, and system design.

**System Development Consultant | [SILEX Consulting Limited](#)**

**July 2025 - Present**

- Led the development of ICCASA's Monitoring and Evaluation system.
- The system includes projects, indicators, outcomes, baselines, and forms for data collection.
- Automatic computations with custom formulae, with a comprehensive reporting system. The system includes tablet mobile apps for data collection by field agents.

**Lead Software Engineer | [AfriLearnTech Ghana Limited](#)**

**September 2025 – Present**

- Led the development of a national digital learning platform for Liberia's Ministry of Education.
- System includes LLM Agents for course content generation to augment localized content.
- System includes (synchronizable) offline boxes to be deployed in schools with low connectivity.

**Software Developer | Iridis Labs****Sept 2023 – Jun 2025**

- Worked on a management software for the Electricity Regulatory Commission of Liberia.
- Recruited and onboarded 10 interns and conducted training sessions for them.
- Technologies used included: Flutter, PostgreSQL, React, TypeScript, and Ubuntu Servers.

**Founder | PKay Software Consultancy****October 2023 – Present**

- Led the development of software systems, including web and mobile, for a wide range of clients.
- Systems include Admission Systems, Management Information Systems, M&E Dashboards, Cross-platform Mobile Applications, and Business Portals.
- Handled stakeholder engagements and the entire Software Development Life Cycle.

**Software Developer | PayHub Ghana Limited****April 2021 - December 2025**

- Developed Payment Processing APIs for internal use and for clients.
  - Developed fintech-grade dashboards for transaction monitoring and management.
  - Reduced system latency through Redis-based caching and optimized session management.
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**PROJECTS****Stockora (Multi-tenant Purchase, Stock and Inventory Management SaaS) | [Link](#)**

- Built Stockora, a multi-tenant SaaS platform for purchase, inventory, and sales management.
- Supports batch-level stock tracking, automated reporting and real-time profitability insights.
- Integrated Paystack subscriptions, email, FCM notifications, and role-based access.
- Includes an offline-capable app, CI/CD deployment, and encrypted backup infrastructure.

**ProCMOS (A Procurement and Contract Management System)**

- Designed and built ProCMOS, a configurable procurement and contract management system.
- Implemented governed workflows for sourcing, evaluations, approvals, supplier management.
- Developed secure institutional role-based access, payments, and comprehensive audit trails.

**ACDI ERP (An Enterprise Resource Planning Software for ACDI)**

- It integrates Finance, HR, Procurement, Inventory, Projects, Training, M&E, and Operations.
- Role-based workflows, approval automation, reporting, and document management.
- Includes an offline app with dynamic field-collection forms and automatic data synchronization.

**Holistic Modern M&E Software as a Service**

- Developed core M&E workflows for projects, outcomes, indicators, versioned forms, field submissions, automated indicator computations, dashboards, and PDF reports.
- The system also includes tablet mobile apps for data collection by field agents.

**Transport and Hospitality Management Software | [Link](#)****July 2026**

- Built a holistic vehicle rental and guest house booking management system.
- System included payment integration, idempotent and atomic transaction processing.
- Comprehensive dashboard for managing and monitoring fleets and guest houses.
- System operates at a 99.9% availability/up time.

**Projection Software | [Link](#)****April 2026**

- Built a desktop church projection platform using Tauri, React, and SQLite.
- Developed a real-time dual-window projection and display engine.
- Engineered offline content management for scriptures, hymns, and announcements.

**AfriLearn Digital Learning Platform | [Link](#)****Oct 2025 - April 2026**

- Built an offline-first national digital learning platform for Liberia's Ministry of Education.
- System includes easy-to-deploy hardware 'boxes' deployable in low connectivity areas.
- Automatic updation and synchronization of educational content (videos, Audio, PDFs, etc.).
- Uses LLM to suggest and generate personalized lessons and assessments for learners.

**Synther | [Link](#)****April 2026**

- A synthetic data generation tool for generating highly configurable datasets
- Implemented Celery workers and Redis for scalable background generation and live progress.
- Designed an intuitive wizard with datatype-aware fields, options, and downloads.
- Ensured data fidelity with numpy/Faker-based generators, and deterministic seeding.

**Nrf24 Firewall FullStack and Embedded | [Link](#)****March – April 2026**

- Designed/implemented a lightweight software-defined firewall for an NRF24-based IP network.
- Built a full-stack system using FastAPI and React with a staged rule engine and web interface.
- Configured Linux TUN interfaces, routing, NAT, and integrated nftables for traffic filtering.
- Evaluated system performance using UDP throughput and latency measurements with iperf3.

**ReviewMate | [Link](#)****November 2025**

- Designed and built a Word add-in with an LLM-powered review agent (TypeScript, Office.js).
- Designed prompt-driven review workflows to generate structured, actionable feedback as review comments in Word.
- Integrated OpenAI-compatible LLM pipeline with JSON outputs and resilient parsing.
- Added AI writing tools (trim, paraphrase, summaries) with secure API key handling.

**Admissions Management System | [Link](#)****January 2024 – Present**

- Designed/developed an admissions management system for [GIMPA](#) (a tertiary institute in Ghana)
- The system includes a management portal for monitoring and management, with an automated and manual notification system.
- Over 2000 applications processed so far, and counting.
- Application processing time is improved by over 500 percent.

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**REFEREES****(1) Prof. Yaokumah Winfred**

Computer Science Department, University of Ghana  
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**(2) Prof. Charles Teye Amoatey**

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Director, SILEX Consulting Limited  
Professor, Ghana Institute of Management and Public Administration  
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**(3) Prof. Kofi Sarpong Adu-Manu**

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